

## Antibiotic use and microbiome function.

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Our microbiome should be understood as one of the most complex components of the human body. The use of  $\beta$ -lactam antibiotics is one of the microbiome covariates that influence its composition. The extent to which our microbiota changes after an antibiotic intervention depends not only on the chemical nature of the antibiotic or cocktail of antibiotics used to treat specific infections, but also on the type of administration, duration and dose, as well as the level of resistance that each microbiota develops. We have begun to appreciate that not all bacteria within our microbiota are vulnerable or reactive to different antibiotic interventions, and that their influence on both microbial composition and metabolism may differ. Antibiotics are being used worldwide on a huge scale and the prescription of antibiotics is continuing to rise; however, their effects on our microbiota have been reported for only a limited number of them. This article presents a critical review of the antibiotics or antibiotic cocktails whose use in humans has been linked to changes in the composition of our microbial communities, with a particular focus on the gut, oral, respiratory, skin and vaginal microbiota, and on their molecular agents (genes, proteins and metabolites). We review the state of the art as of June 2016, and cover a total of circa 68 different antibiotics. The data herein are the first to compile information about the bacteria, fungi, archaea and viruses most influenced by the main antibiotic treatments prescribed nowadays.

## Development of SYN-004, an oral beta-lactamase treatment to protect the gut microbiome from antibiotic-mediated damage and prevent Clostridium difficile infection.

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The gut microbiome, composed of the microflora that inhabit the gastrointestinal tract and their genomes, make up a complex ecosystem that can be disrupted by antibiotic use. The ensuing dysbiosis is conducive to the emergence of opportunistic pathogens such as Clostridium difficile. A novel approach to protect the microbiome from antibiotic-mediated dysbiosis is the use of beta-lactamase enzymes to degrade residual antibiotics in the gastrointestinal tract before the microflora are harmed. Here we present the preclinical development and early clinical studies of the beta-lactamase enzymes, P3A, currently referred to as SYN-004, and its precursor, P1A. Both P1A and SYN-004 were designed as orally-delivered, non-systemically available therapeutics for use with intravenous beta-lactam antibiotics. SYN-004 was engineered from P1A, a beta-lactamase isolated from Bacillus licheniformis, to broaden its antibiotic degradation profile. SYN-004 efficiently hydrolyses penicillins and cephalosporins, the most widely used IV beta-lactam antibiotics. In animal studies, SYN-004 degraded ceftriaxone in the GI tract of dogs and protected the microbiome of pigs from ceftriaxone-induced changes. Phase I clinical studies demonstrated SYN-004 safety and tolerability. Phase 2 studies are in progress to assess the utility of SYN-004 for the prevention of antibiotic-associated diarrhea and Clostridium difficile disease.

## SYN-007, an Orally Administered Beta-Lactamase Enzyme, Protects the Gut Microbiome from Oral Amoxicillin/Clavulanate without Adversely Affecting Antibiotic Systemic Absorption in Dogs.

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Beta-lactamases, enzymes produced by bacteria to degrade beta-lactam antibiotics, have been harnessed as therapeutics to protect the gut microbiome from damage caused by antibiotics. Proof-of-concept of this approach using SYN-004 (ribaxamase), a beta-lactamase formulated for oral delivery with intravenous (IV) penicillins and cephalosporins, was demonstrated with animal models and in humans. Ribaxamase degraded ceftriaxone in the gastrointestinal tract, protected the gut microbiome, significantly reduced the incidence of *Clostridioides difficile* disease and attenuated emergence of antibiotic resistant organisms. SYN-007 is a delayed release formulation of ribaxamase intended for use with oral beta-lactams. In dogs treated with oral amoxicillin, SYN-007 diminished antibiotic-mediated microbiome disruption and reduced the emergence of antibiotic resistance without altering amoxicillin systemic absorption. Here, SYN-007 function in the presence of clavulanate, a beta-lactamase inhibitor, was investigated. Dogs received amoxicillin (40 mg/kg, orally (PO), three times a day (TID)) or the combined antibiotic/beta-lactamase inhibitor, amoxicillin/clavulanate (40 mg/kg amoxicillin, 5.7 mg/kg clavulanate, PO, TID) +/- SYN-007 (10 mg, PO, TID) for five days. Serum amoxicillin levels were not significantly different +/- SYN-007 compared to amoxicillin alone or amoxicillin/clavulanate alone as controls for both first and last doses, indicating SYN-007 did not interfere with systemic absorption of the antibiotic. Whole genome shotgun metagenomics analyses of the fecal microbiomes demonstrated both amoxicillin and amoxicillin/clavulanate significantly reduced diversity and increased the frequency of antibiotic resistance genes. Microbiome damage appeared more severe with amoxicillin/clavulanate. In contrast, with SYN-007, microbiome diversity was not significantly altered, and frequency of antibiotic resistance genes did not increase. Importantly, SYN-007 functioned in the presence of clavulanate to protect the gut microbiome indicating that SYN-007 activity was not inhibited by clavulanate in the dog gastrointestinal tract. SYN-007 has the potential to expand microbiome protection to beta-lactam/beta-lactamase inhibitor combinations delivered orally or systemically.

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## Oral Beta-Lactamase Protects the Canine Gut Microbiome from Oral Amoxicillin-Mediated Damage.

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Antibiotics damage the gut microbiome, which can result in overgrowth of pathogenic microorganisms and emergence of antibiotic resistance. Inactivation of antibiotics in the small intestine represents a novel strategy to protect the colonic microbiota. SYN-004 (ribaxamase) is a beta-lactamase formulated for oral delivery intended to degrade intravenously administered beta-lactam antibiotics in the gastrointestinal (GI) tract. The enteric coating of ribaxamase protects the enzyme from stomach acid and mediates pH-dependent release in the upper small intestine, the site of antibiotic biliary excretion. Clinical benefit was established in animal and human studies in which ribaxamase was shown to degrade ceftriaxone in the GI tract, thereby preserving the gut microbiome, significantly reducing *Clostridioides difficile* disease, and attenuating antibiotic resistance. To expand ribaxamase utility to oral beta-lactams, delayed release formulations of ribaxamase, SYN-007, were engineered to allow enzyme release in the lower small intestine, distal to the site of oral antibiotic absorption. Based on in vitro dissolution profiles, three SYN-007 formulations were selected for evaluation in a canine model of antibiotic-mediated gut dysbiosis. Dogs received amoxicillin (40 mg/kg, PO, TID) +/- SYN-007 (10 mg, PO, TID) for five days. Serum amoxicillin levels were measured after the first and last antibiotic doses and gut microbiomes were evaluated using whole genome shotgun sequence metagenomics analyses of fecal DNA prior to and after antibiotic treatment. Serum amoxicillin levels did not significantly differ +/- SYN-007 after the first

dose for all SYN-007 formulations, while only one SYN-007 formulation did not significantly reduce systemic antibiotic concentrations after the last dose. Gut microbiomes of animals receiving amoxicillin alone displayed significant loss of diversity and emergence of antibiotic resistance genes. In contrast, for animals receiving amoxicillin + SYN-007, microbiome diversities were not altered significantly and the presence of antibiotic resistance genes was reduced. These data demonstrate that SYN-007 diminishes amoxicillin-mediated microbiome disruption and mitigates emergence and propagation of antibiotic resistance genes without interfering with antibiotic systemic absorption. Thus, SYN-007 has the potential to protect the gut microbiome by inactivation of beta-lactam antibiotics when administered by both oral and parenteral routes and to reduce emergence of antibiotic-resistant pathogens.

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## Reducing Broad-Spectrum Antibiotic Treatment of Simple Group A Streptococcal Infections to Reduce Harm to the Microbiome.

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**Background** Broad-spectrum antibiotics disrupt the human microbiome resulting in a greater risk of harmful, long-term conditions that impact human health. Group A streptococcal (GAS) infections can be treated with penicillin. **Objective** We examined the treatment of simple GAS infections to assess the use of broad-spectrum antibiotics. **Methods** Smart relational database extraction queries from January 1, 2016 to July 10, 2019 (3.6 years) of patients less than 22 years old in a 4-hospital system electronic medical record (EMR). **Results** We found 1778 non-ED outpatients and 873 ED patients with simple GAS infections who were not allergic to penicillin. A total of 75% and 44% of non-ED and ED patients were treated with broad-spectrum antibiotics, respectively ( $p < 0.001$ ). Older patients were treated with penicillin alone more frequently than younger age groups ( $p < 0.001$ ). **Conclusion** These findings highlight opportunities for clinicians to reduce the utilization of broad-spectrum antibiotics for the treatment of simple GAS infections to reduce harm to the microbiome.

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## Development and validation of a liquid chromatography high-resolution mass spectrometry orbitrap method for the sensitive quantification of amoxicillin, piperacillin, tazobactam and meropenem in human faeces.

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Beta-lactam antibiotics are of vital importance for the treatment of infections in a broad range of patients. Although most systemically administered antibiotics will be excreted renally, a fraction will reach the gastro-intestinal tract, affecting the intestinal microbiome by eradicating a wide range of bacterial species while facilitating the growth of antimicrobial-resistant species. A better understanding of the kinetics of beta-lactam antibiotics in the gastro-intestinal tract is essential to study their role in the development of antibiotic resistance in bacteria and to help develop future therapies to prevent damage to, or restore, the intestinal microbiome. Analysis of beta-lactam antibiotics in faeces is particularly challenging due to the heterogeneous nature of the matrix, rapid degradation of some beta-lactam antibiotics in faeces and very strong ion suppression when using mass spectrometry. Sample preparation was optimized using a sequential strategy of experimental designs. It resulted in lyophilization, a MOPS buffer system and the addition of the beta-lactamase inhibitor avibactam to minimize degradation of antibiotics allowing sensitive quantification. The developed liquid chromatography method with

high-resolution mass spectrometric detection was successfully validated according to bioanalytical EMA guidelines and had a linear range of 1-200 µg g<sup>-1</sup> lyophilized faeces for amoxicillin, piperacillin and meropenem; and 0.5-100 µg g<sup>-1</sup> lyophilized faeces for tazobactam. Despite the highly complex and heterogeneous composition of faeces, the accuracy (0.1-15%) and precision (1.7-12.1%) were in line with those obtained for quantification methods of beta-lactam antibiotics in plasma, the golden standard matrix for therapeutic drug monitoring. The applicability of the method was illustrated by successful quantification of piperacillin and tazobactam in faeces from an intensive care unit patient receiving piperacillin/tazobactam in a continuous intravenous infusion. Both piperacillin and tazobactam were still present six days after discontinuation of the therapy.

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## Oral Metallo-Beta-Lactamase Protects the Gut Microbiome From Carbapenem-Mediated Damage and Reduces Propagation of Antibiotic Resistance in Pigs.

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Antibiotics can damage the gut microbiome, leading to serious adventitious infections and emergence of antibiotic resistant pathogens. Antibiotic inactivation in the GI tract represents a strategy to protect colonic microbiota integrity and reduce antibiotic resistance. Clinical utility of this approach was established when SYN-004 (ribaxamase), an orally-administered beta-lactamase, was demonstrated to degrade ceftriaxone in the GI tract and preserve the gut microbiome. Ribaxamase degrades penicillins and cephalosporin beta-lactams, but not carbapenems. To expand this prophylactic approach to include all classes of beta-lactam antibiotics, a novel carbapenemase, formulated for oral administration, SYN-006, was evaluated in a porcine model of antibiotic-mediated gut dysbiosis. Pigs (20 kg, n = 16) were treated with the carbapenem, ertapenem (ERT), (IV, 30 mg/kg, SID) for 4 days and a cohort (n = 8) also received SYN-006 (PO, 50 mg, QID), beginning the day before antibiotic administration. ERT serum levels were not statistically different in ERT and ERT + SYN-006 groups, indicating that SYN-006 did not alter systemic antibiotic levels. Microbiomes were evaluated using whole genome shotgun metagenomics analyses of fecal DNA collected prior to and after antibiotic treatment. ERT caused significant changes to the gut microbiome that were mitigated in the presence of SYN-006. In addition, SYN-006 attenuated emergence of antibiotic resistance, including encoded beta-lactamases and genes conferring resistance to a broad range of antibiotics such as aminoglycosides and macrolides. SYN-006 has the potential to become the first therapy designed to protect the gut microbiome from all classes of beta-lactam antibiotics and reduce emergence of carbapenem-resistant pathogens.

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## Identification, Characterization, and Formulation of a Novel Carbapenemase Intended to Prevent Antibiotic-Mediated Gut Dysbiosis.

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Antibiotics can damage the gut microbiome leading to opportunistic infections and the emergence of antibiotic resistance. Microbiome protection via antibiotic inactivation in the gastrointestinal (GI) tract represents a strategy to limit antibiotic exposure of the colonic microbiota. Proof of concept for this approach was achieved with an orally-administered beta-lactamase enzyme, SYN-004 (ribaxamase), that was demonstrated to degrade ceftriaxone excreted into the GI tract and protect the gut microbiome from antibiotic-mediated dysbiosis. Ribaxamase efficiently degrades penicillin and cephalosporin beta-lactam

antibiotics, but is not active against carbapenems. To expand this microbiome protection strategy to include all classes of beta-lactams, three distinct carbapenemases were evaluated for manufacturability, antibiotic degradation spectrum, and stability in human intestinal fluid. *E. coli* production strains were generated for P2A, a novel metallo-enzyme isolated from *B. cereus*, New Delhi metallo-beta-lactamase (NDM), and *Klebsiella pneumoniae* carbapenemase (KPC). While all three enzymes effectively inactivated a broad range of antibiotics, including penicillins, most cephalosporins, and carbapenems in vitro, only P2A retained biological activity when incubated with human chyme. As functional stability in the intestinal tract is a key requirement for an orally-delivered enzyme, P2A was chosen as a potential clinical candidate. An enteric formulation of P2A was developed, called SYN-006, that was inert under high acid conditions, with enzyme dissolution occurring at pH > 5.5. SYN-006 has the potential to expand microbiome protection via antibiotic inactivation to include all classes of beta-lactam antibiotics.

## Metabolic changes associated with polysaccharide utilization reduce susceptibility to some $\beta$ -lactams in *Bacteroides thetaiotaomicron*.

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Antibiotic therapy alters bacterial abundance and metabolism in the gut microbiome, leading to dysbiosis and opportunistic infections. *Bacteroides thetaiotaomicron* (Bth) is both a commensal in the gut and an opportunistic pathogen in other body sites. Past work has shown that Bth responds to  $\beta$ -lactam treatment differently depending on the metabolic environment both in vitro and in vivo. Studies of other bacteria show that an increase in respiratory metabolism independent of growth rate promotes susceptibility to bactericidal antibiotics. We propose that Bth enters a protected state linked to an increase in polysaccharide utilization and a decrease in the use of simple sugars. Here, we apply antibiotic susceptibility testing, transcriptomic analysis, and genetic manipulation to characterize this polysaccharide-mediated tolerance (PM tolerance) phenotype. We found that a variety of mono- and disaccharides increased the susceptibility of Bth to several different  $\beta$ -lactams compared to polysaccharides. Transcriptomics indicated a metabolic shift from reductive to oxidative branches of the tricarboxylic acid cycle on polysaccharides. Accordingly, supplementation with intermediates of central carbon metabolism had varying effects on PM tolerance. Transcriptional analysis also showed a decrease in the expression of the electron transport chain (ETC) protein NQR and an increase in the ETC protein NUO, when given fiber versus glucose. Deletion of NQR increased Bth susceptibility while deletion of NUO and a third ETC protein NDH2 had no effect. This work confirms that carbon source utilization modulates antibiotic susceptibility in Bth and that anaerobic respiratory metabolism and the ETC play an essential role. **IMPORTANCE** Antibiotics are indispensable medications that revolutionized modern medicine. However, their effectiveness is challenged by a large array of resistance and tolerance mechanisms. Treatment with antibiotics also disrupts the gut microbiome which can adversely affect health. *Bacteroides* are prevalent in the gut microbiome and yet are frequently involved in anaerobic infections. Thus, understanding how antibiotics affect these bacteria is necessary to implement proper treatment. Recent work has investigated the role of metabolism in antibiotic susceptibility in distantly related bacteria such as *Escherichia coli*. Using antibiotic susceptibility testing, transcriptomics, and genetic manipulation, we demonstrate that polysaccharides reduce  $\beta$ -lactam susceptibility when compared to monosaccharides. This finding underscores the profound impact of metabolic adaptation on the therapeutic efficacy of antibiotics. In the long term, this work indicates that modulation of metabolism could make *Bacteroides* more susceptible during infections or protect them in the context of the microbiome.

## An engineered live biotherapeutic for the prevention of antibiotic-induced dysbiosis.

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Antibiotic-induced alterations in the gut microbiota are implicated in many metabolic and inflammatory diseases, increase the risk of secondary infections and contribute to the emergence of antimicrobial resistance. Here we report the design and in vivo performance of an engineered strain of *Lactococcus lactis* that altruistically degrades the widely used broad-spectrum antibiotics  $\beta$ -lactams (which disrupt commensal bacteria in the gut) through the secretion and extracellular assembly of a heterodimeric  $\beta$ -lactamase. The engineered  $\beta$ -lactamase-expression system does not confer  $\beta$ -lactam resistance to the producer cell, and is encoded via a genetically unlinked two-gene biosynthesis strategy that is not susceptible to dissemination by horizontal gene transfer. In a mouse model of parenteral ampicillin treatment, oral supplementation with the engineered live biotherapeutic minimized gut dysbiosis without affecting the ampicillin concentration in serum, precluded the enrichment of antimicrobial resistance genes in the gut microbiome and prevented the loss of colonization resistance against *Clostridioides difficile*. Engineered live biotherapeutics that safely degrade antibiotics in the gut may represent a suitable strategy for the prevention of dysbiosis and its associated pathologies.

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